

Geography Ch-10 (Must-Do) — Coastal Landforms

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain
Wave erosion & deposition; emergent vs submergent coasts (ria, fjord, Dalmatian)
CA anchor: India coastal erosion (NCCR, 33.6% of coast)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Coastal Erosion, Deposition & Coast Types (GC Leong / Majid)

■ NEWS TRIGGER

Coastal geomorphology explains India's 7,500-km shoreline — its cliffs, beaches, spits and lagoons — now under stress as one-third of the coast erodes.

■ INTRO & ORIGIN

Waves erode by **hydraulic action**, **abrasion (corrasion)**, **attrition and solution**, and redistribute sediment by **longshore drift** to build depositional forms. Coasts are also classed as emergent or submergent.

Wave attack on cliff bases cuts notches -> caves -> arches -> stacks -> stumps, while drifted sand builds beaches, spits, bars and tombolos. Sea-level/land movement gives emergent or submergent coasts.

■ KEY DATA

Group	Landform	What it is
Erosional	Cliff & wave-cut platform	Steep face + gentle bench that widens as cliff retreats
Erosional	Cave -> Arch -> Stack -> Stump	Sequence from headland wave attack (e.g. Needles)
Erosional	Geo / Blowhole	Cleft / vertical hole where a cave roof collapses
Depositional	Beach / Spit / Bar	Sediment ridge / spit across bay / bar enclosing lagoon
Depositional	Tombolo	Spit that ties an island to the mainland
Submergent coasts	Ria / Fjord / Dalmatian	Drowned valley / glacial trough / parallel-ridge coast

■ STATIC THEORY

- **Erosional sequence:** wave attack cuts **caves** in a headland; two caves meet to form an **arch**; the arch collapses leaving a **stack**, then a **stump** (Majid & GC Leong).
- **Wave-cut platform:** gently seaward-sloping bench at the cliff base that **widens as the cliff retreats**.
- **Depositional forms** built by **longshore drift**: **beaches**, **spits**, **bars** (enclosing lagoons) and **tombolos** (island tied to shore).
- **Submergent coasts:** **ria** (drowned river valley), **fjord** (drowned glacial trough), **Dalmatian** (drowned ridges parallel to coast), estuarine coast.
- **Emergent coasts:** uplift produces **raised beaches** and straight cliffed coasts with deeper offshore water (GC Leong).

■ MUST-KNOW FACTS

1. Erosion sequence: cave -> arch -> stack -> stump.
2. Wave-cut platform widens as the cliff retreats.

■ UPSC TRAPS

- ✗ A fjord is a drowned river valley.
- ✓ A fjord is a drowned GLACIAL trough; a drowned RIVER valley is a ria.

3. Longshore drift builds spits, bars, tombolos. 4. Tombolo ties an island to the mainland. 5. Submergent: ria (river), fjord (glacier), Dalmatian (parallel ridges). 6. Emergent coast -> raised beach, straight cliffs.	✗ A tombolo separates an island from the mainland. ✓ A tombolo CONNECTS an island to the mainland (a linking spit). ✗ A stack forms before an arch. ✓ Order is cave -> arch -> stack -> stump (stack forms AFTER arch collapse).
■ MAINS ANGLE <i>Explain wave erosion vs deposition and emergent/submergent coasts, applying them to India's straight submergent west coast and deltaic emergent east coast.</i>	■ STUDY LINK Geography → Physical → Coastal Landforms (GC Leong: Coasts; Majid: Coastal Geomorphology)

■ MEDIUM UPSC RELEVANCE

MEDIUM CA ANCHOR — India's Coastal Erosion (NCCR)	GS-I / GS-III · Wave erosion applied → Shoreline management
■ NEWS TRIGGER The National Centre for Coastal Research (NCCR) reports ~33.6% of India's mainland coast is eroding (West Bengal ~63%, Puducherry ~57%, Kerala ~45%, Tamil Nadu ~41%).	
■ INTRO & ORIGIN Coastal erosion is wave attack outpacing sediment supply — accelerated by sea-level rise, dams cutting sediment, mangrove loss and hard structures that disrupt longshore drift.	
■ STATIC THEORY - NCCR mapping: ~33.6% eroding, ~26.9% accreting, ~39.6% stable. - Drivers: sea-level rise, cyclones, reduced sediment (river dams), mangrove loss, and coastal structures (ports/seawalls) that interrupt longshore drift. - Responses: shoreline management plans, beach nourishment, mangrove/bio-shields.	
■ MUST-KNOW FACTS 1. NCCR: ~33.6% of India's coast eroding. 2. Worst: West Bengal (~63%), Puducherry (~57%). 3. Cause link: sea-level rise + sediment starvation + hard structures.	■ UPSC TRAPS ✗ Seawalls always stop coastal erosion everywhere. ✓ Hard structures often shift erosion downdrift by disrupting longshore drift.
■ MAINS ANGLE <i>Link wave-erosion theory to India's shoreline crisis and integrated coastal-zone management.</i>	■ STUDY LINK Geography → Coasts → Coastal Erosion (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI